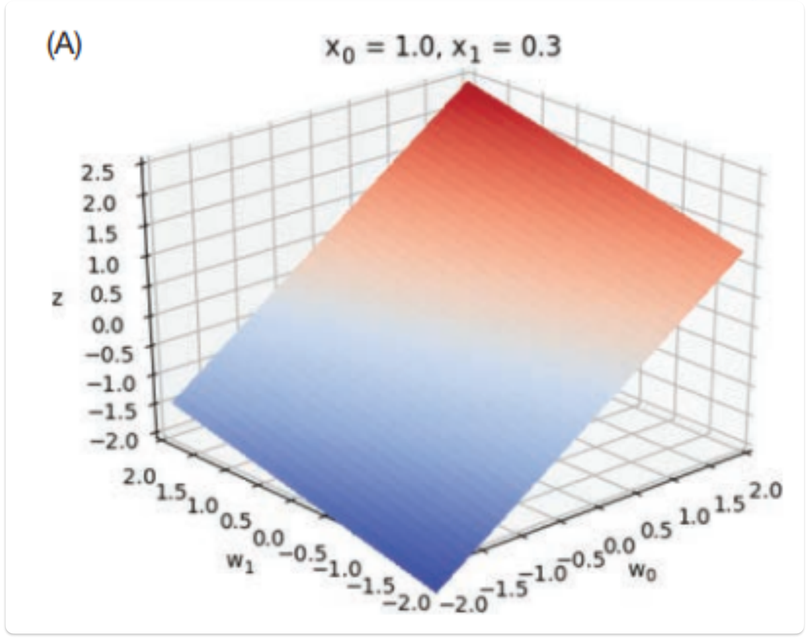
1. Adjustment of Weight in 3D plot:

A. for one input perceptron, given a specific pair training example { inputs ($x_0=1$ & $x_1=0.3$) | output $y=-1$ },
so that we can observe perceptron learning algorithm adjusting weights (x_0, x_1) in different value,
and corresponding z and predicted output P_y will change so as to match desired output y (ground truth).

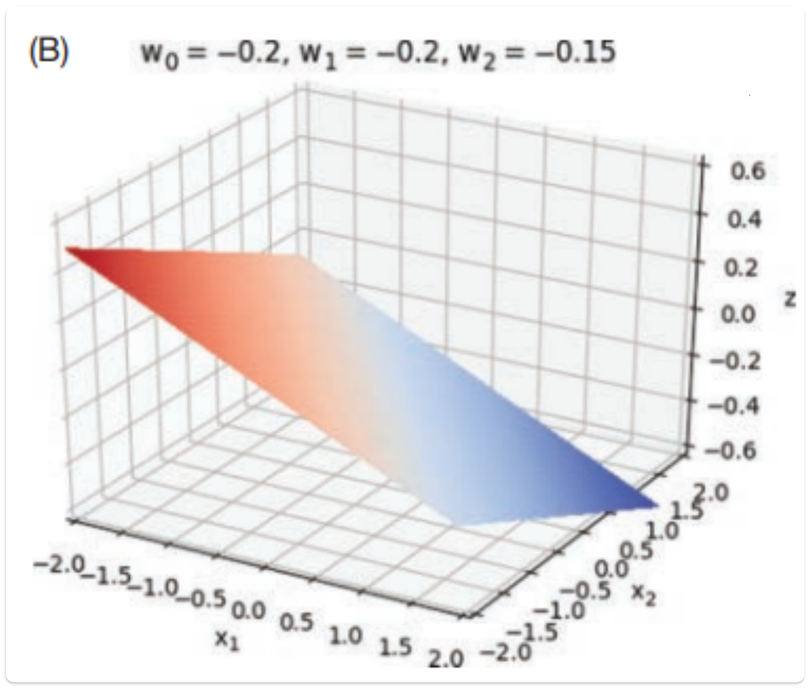
B.



2. Variability between Different set of Inputs in 3D plot:

A. for two input perceptron, given a specific set of weights ($w_0=-0.2, w_1=-0.2, w_2=-0.15$),
so that we can understand how z will change when different sets of inputs (x_1, x_2) put into two-input perceptron.

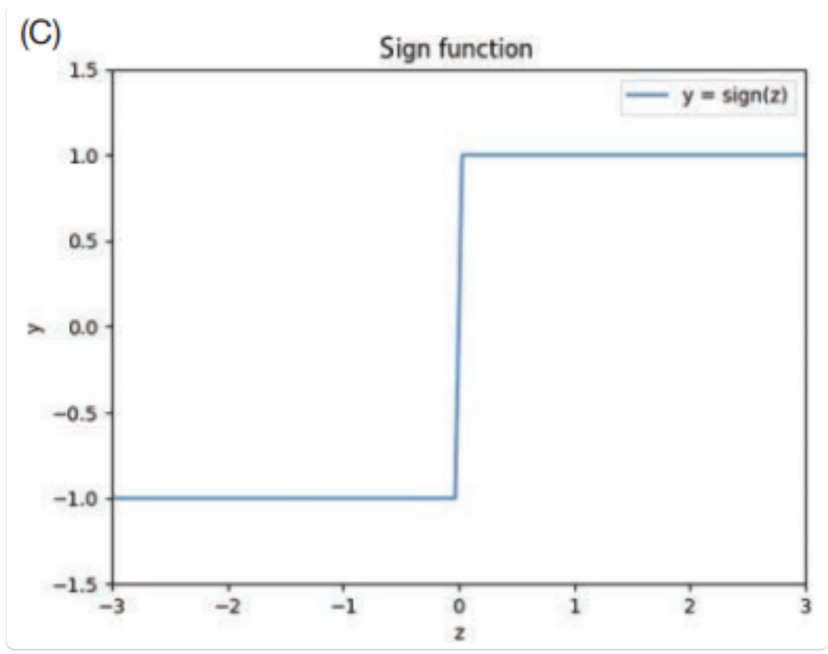
B.



3. Weighted Sum Function & Activation Function in 2D plot:

A. For all perceptron regardless of their statement of weights and input,
simply show the mutual relationship between predicted output P_y and corresponding z

B.



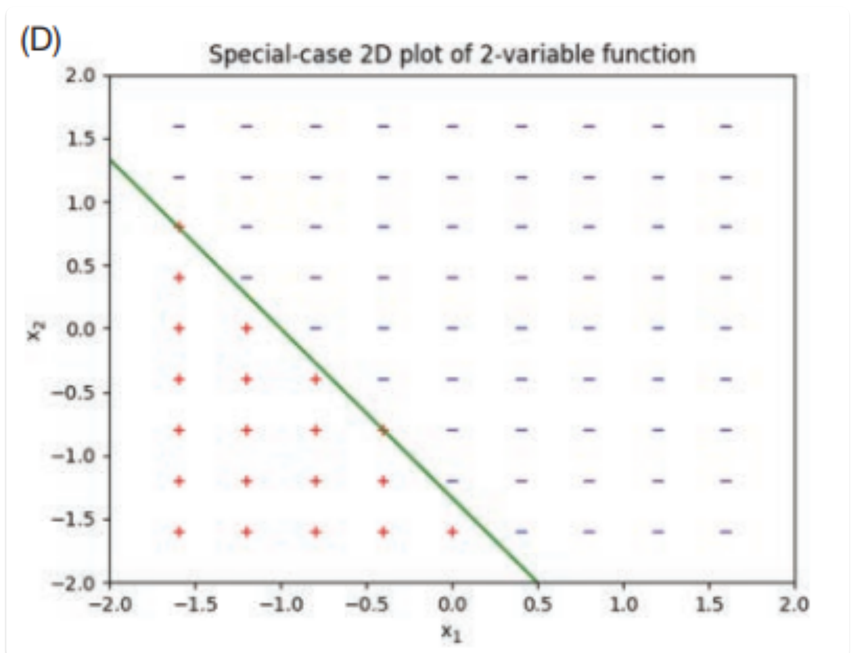
4. Decision Boundry Equation in 2D plot:

A. for two input perceptron, given a specific set of weights ($w_0=-0.2, w_1=-0.2, w_2=-0.15$),
by putting different pair training examples { $x_0=1, x_1=?, x_2=?$ | output y (ground truth) $= +1$ or -1 } on the chart,

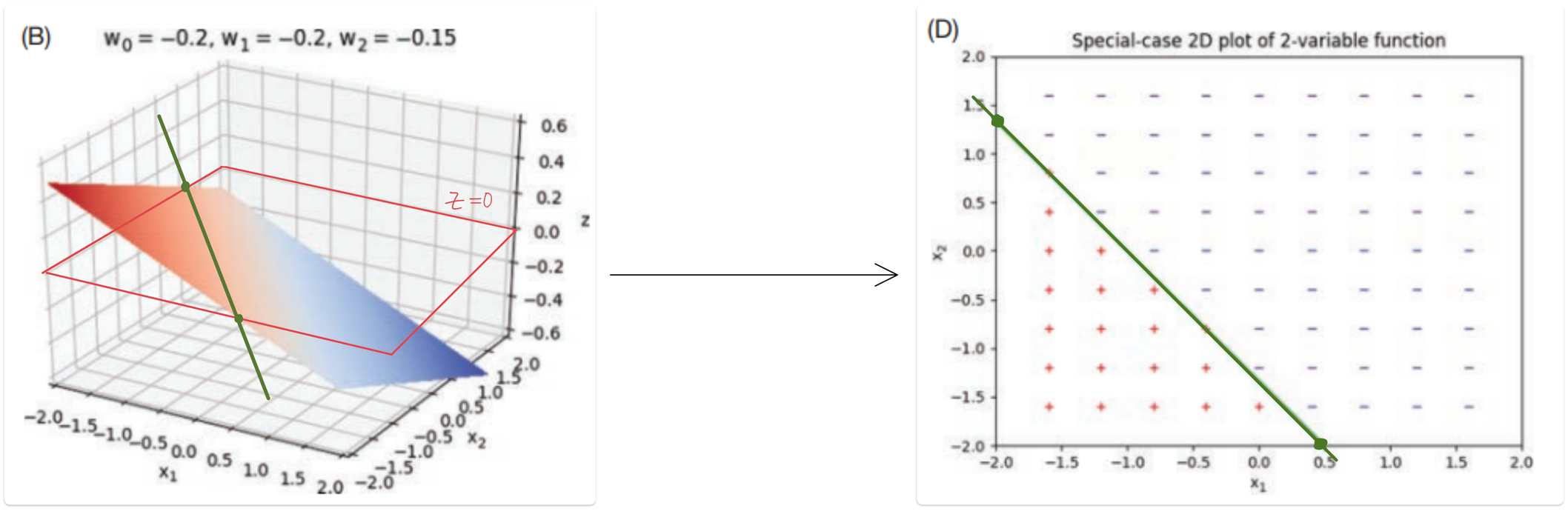
We can use the straight line: $w_0 x_0 + w_1 x_1 + w_2 x_2 = 0 \longrightarrow x_2 = -\frac{w_1}{w_2} x_1 - \frac{w_0}{w_2}$ (slope of line; y-intercept of line) to divide the graph into 3 parts. $\Longrightarrow$

Part1. on the line: $w_0 x_0 + w_1 x_1 + w_2 x_2 = 0$
Part2. one side of the line (the side must be the area full of output=⊕): $w_0 x_0 + w_1 x_1 + w_2 x_2 > 0$
Part3. the other side of the line (the side must be the area full of output=⊖): $w_0 x_0 + w_1 x_1 + w_2 x_2 < 0$

B



C. looking at Variability between Different set of Inputs in 3D plot from above,
and draw the intersection line with $z=0$, we can get Decision Boundry Equation in 2D plot



D. this type of plot is very common when looking at any binary classification problem.